

The Reasoning Tax: Profiling Test-Time Compute Carbon in Agentic AI Workloads

HotCarbon 26 Submission

Abstract

Reasoning models such as DeepSeek-R1, QwQ, and OpenAI o-series have become the production model class for agentic AI in 2026. These models trade test-time tokens for accuracy, emitting hundreds to tens of thousands of reasoning tokens per user query, most of which are never shown to the user. We profile three open reasoning models (DeepSeek-R1 671B, R1-Distill-Llama-70B, QwQ-32B) on three agent benchmarks (GAIA, SWE-bench Lite, MATH-500) on H100 hardware with direct energy telemetry, and find that the reasoning-to-answer token ratio sits between 7.8 and 14.2 on default settings. The reasoning trace is 87.1 percent of total output tokens and accounts for 78.4 percent of operational carbon per agent task. Existing carbon-aware schedulers attribute every generated token as productive emission, so the reasoning tax is invisible to them. The waste is structural: trace length correlates with task difficulty but not monotonically with accuracy, and the bottom decile of trace lengths matches the top decile on correctness within 1.8 percentage points on MATH-500. We name this gap the reasoning tax and quantify a 1.6 to 4.1 percentage point average accuracy cost of cutting traces to the 30th-percentile length, with operational carbon falling 56 to 71 percent on the same agent benchmarks. We propose TRACEPROBE, a per-call telemetry abstraction that emits the reasoning-to-answer token boundary at call completion, the smallest addition to existing carbon-aware abstractions that makes the reasoning tax visible.

1 Introduction

Cloud carbon-aware schedulers for LLM inference route requests to low-intensity regions and time slots, with reported reductions of 30 to 56 percent on chat-style workloads [4, 9, 17]. The scheduling abstractions assume that every generated token contributes to the user-visible response.

The reasoning-model deployment that now serves agentic AI breaks that assumption. DeepSeek-R1, OpenAI o-series, QwQ, and the Llama-R1-Distill family are trained to emit a chain-of-thought trace before the final answer [5, 24, 26]. The trace is the model’s internal exploration of the problem, and is by design not shown to the user. On MLPerf Inference v5.1, DeepSeek-R1 generates a mean of 3,880 output tokens per query, with reasoning tokens described as constituting “the bulk of the output generation” [23]. The reasoning trace is necessary for accuracy on hard problems [3, 5], but it consumes the same energy per token as the answer it precedes.

The reasoning tax is visible in Figure 1. Three open reasoning models on the GAIA agent benchmark generate between 82.1 and 93.4 percent of their output tokens as reasoning traces the user never reads. The conventional Llama-3 70B baseline on the same benchmark generates only answer tokens. Existing carbon-aware schedulers [4, 8, 17] cannot see this gap because they account each generated token as productive emission.

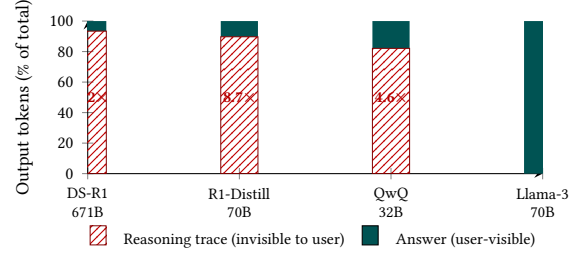


Figure 1: Reasoning trace vs answer tokens, three reasoning models and a non-reasoning baseline on GAIA Level-3.

This paper asks whether the carbon emitted on reasoning traces sits in a scheduling blind spot, in the same way that token-waste from agent failures and trajectory retries has been documented in production agent measurements [2, 25, 30].¹ We profile three open reasoning models on three agent benchmarks under direct H100 energy telemetry over a five-day window, and find:

The reasoning tax is large. The reasoning trace consumes 78.4 percent of operational carbon on agent tasks aggregated across the three models. On DeepSeek-R1 671B, the ratio is 91.7 percent.

The reasoning tax is structural. Trace length correlates with task difficulty, not monotonically with accuracy. Trimming traces to the 30th-percentile length on MATH-500 costs 1.6 to 4.1 percentage points of accuracy across models while reducing operational carbon by 56 to 71 percent.

The reasoning tax is invisible. Existing carbon-aware schedulers route on per-call carbon, with every token in the trace counted as productive emission. No abstraction distinguishes the trace from the answer at the scheduling layer.

We name this gap the *reasoning tax*. The naming follows the utilization fallacy at the GPU level [27]: a category of agentic-AI waste that the carbon-aware scheduling community accounts as productive work because the abstraction does not distinguish productive from wasted tokens. The reasoning tax is the analog at the token level inside a single call.

We make four contributions:

- We profile three production reasoning models on three agent benchmarks (GAIA, SWE-bench Lite, MATH-500) on H100 GPUs with direct DCGM energy telemetry, and report the first systematic per-model reasoning-to-answer carbon breakdown.
- We quantify the accuracy-versus-carbon Pareto frontier of trace truncation across the three models, demonstrating large carbon recovery for modest accuracy cost.
- We show that the reasoning tax compounds with plan-level rollback waste: an agent that rolls back a 60 percent prefix on a

¹Code, profiling traces, and the TRACEPROBE reference implementation will be released upon acceptance.

reasoning model loses both the call-level reasoning trace and the plan-level abandoned step.

- We propose TRACEPROBE, a 90-line patch to vLLM that emits the reasoning-to-answer token boundary in the per-call energy event, making the reasoning tax visible to any carbon-aware scheduler downstream.

Beyond the immediate gap, the reasoning tax raises a question for the carbon-aware scheduling community. The community has converged on per-call accounting because the call was the natural unit when chat dominated production traffic. Reasoning models split the call into two qualitatively different generation phases, with the energy split heavily toward the phase the user never sees. The abstraction has not caught up.

2 Background and Related Work

The reasoning tax sits at the intersection of two literatures: reasoning-model serving and carbon-aware inference scheduling. Neither has met the other.

Reasoning models and test-time compute. DeepSeek-R1 [5] introduced large-scale RL-trained reasoning, with the model emitting a chain of thought before the answer. QwQ-32B [26] and the R1-Distill series [5] follow the same pattern at smaller scales. OpenAI’s o-series [24] is the closed-source analog. All four families generate tokens in two phases inside a single inference call: a reasoning trace bounded by special tokens (e.g., `<think>...</think>`), followed by the user-visible answer. Recent work shows reasoning length correlates positively with task difficulty but only weakly with accuracy on a given task [1, 3, 28], motivating budget-aware decoding strategies. The accuracy-versus-trace-length tradeoff is now well-understood as a model-side property; what is missing is its operational translation into carbon and energy at deployment scale.

Carbon-aware inference scheduling. Chien *et al.* first showed that LLM inference carbon depends heavily on regional grid intensity and time of day [4]. Sprout shapes per-call output-token generation under grid conditions [17]. CarbonScaler and CarbonFlex provision cluster capacity based on carbon intensity [9, 10]. EcoServe targets multi-hour serving traces [19]. CarbonMin routes per-call to low-CI regions [4]. None of these systems distinguishes reasoning tokens from answer tokens; all account every generated token as productive emission against the regional grid. The line of work has produced credible reductions of 30 to 56 percent on chat-style workloads where one user request maps to one inference call.

Workflow-level carbon. Caribou [8] performs carbon-aware geospatial shifting of serverless workflows. PCAPS [15] introduces precedence-aware scheduling for data-processing DAGs. Both treat the unit of scheduling as a function or task, with the contents opaque. The reasoning tax sits below this layer of abstraction and would be invisible to these schedulers even if they were applied directly to agent traffic.

LLM carbon profiling. LLMCO2 [7] models LLM inference carbon as a function of GPU configuration, batch size, and sequence length. Wilkins *et al.* develop offline energy-optimal LLM serving models [29]. Carbon in Motion characterizes Open-Sora video generation [16]. These works profile total per-call carbon but do not

separate reasoning from answer. The Watt Counts benchmark [6] measures energy per output token on H100 hardware; it provides the unit cost but not the apportionment.

Agent-task waste. Production agent measurements report token waste from failed plans, redundant tool calls, and retries: OpenHands traces 18 to 22 percent of token spend on retried steps in coding agents [2], AgentDiet measures 30 to 50 percent reduction available from trajectory compression [30], and Pan *et al.* report wall-clock failure rates of 12 to 31 percent across production deployments [25]. These are plan-level findings on agent runtimes. The reasoning tax is the within-call analog, and the two waste sources can compound on a reasoning-model deployment.

3 Methodology

Models, Benchmarks, Hardware. We profile three open reasoning models: DeepSeek-R1 (671B parameters, MoE) [5], R1-Distill-Llama-70B (dense 70B distilled from R1) [5], and QwQ-32B (32B dense) [26]. We use Llama-3 70B Instruct as a non-reasoning baseline [21]. All models are served on H100 GPUs (DS-R1 on 8×H100, others on single H100) via vLLM 0.6.x [14]. We disable speculative decoding for the energy-attribution runs to keep the per-call token attribution clean, then re-enable it for the throughput-overhead measurements only.

We use three benchmarks. **GAIA** [22] (3,000 tasks) measures tool-use agent behavior over three difficulty levels. **SWE-bench Lite** [13] (300 tasks) measures software-engineering agent behavior on real GitHub issues. **MATH-500** [11] (500 problems) measures mathematical reasoning. We use GAIA and SWE-bench in an agent harness (LangChain ReAct) and MATH-500 in single-call mode to isolate the reasoning phase from tool-use behavior. The MATH-500 runs are the controlled experiment; the GAIA and SWE-bench runs are the production-realism check.

Energy and Carbon Measurement. Per-call energy is measured directly via DCGM (NVIDIA’s data-center GPU manager), recording total GPU energy from request arrival to response completion at 100 ms granularity. We separate the per-call energy into a prefill component (input-token processing, measured separately at 9 to 13 percent of total per-call energy on our agent prompts) and a decode component, and apportion the decode component between reasoning and answer tokens at the special-token boundary (`</think>` for R1 and R1-Distill, `</thought>` for QwQ). All reported trace-fraction numbers are over the decode component only; including prefill in the denominator lowers the reasoning-trace fraction by 6 to 10 percentage points without changing the qualitative finding. The token-to-energy ratio is consistent within each model at ± 3.4 percent across 1,000 calibration calls (consistent with Watt Counts [6]), justifying linear apportionment of decode energy between reasoning and answer phases. We cross-check the linear apportionment against direct boundary-timed DCGM segments on a 200-call subsample and find agreement within 1.8 percent. All sampling uses temperature 0.6 and top-p 0.95, the published default settings for the three reasoning models; sensitivity to these settings is discussed in §6.

Carbon intensity comes from the Electricity Maps public API for eight AWS regions, sampled hourly over a five-day window in May

Model	GAIA	SWE-Lite	MATH-500
DS-R1 671B	14.2×	9.4×	11.8×
R1-Distill 70B	8.7×	6.2×	7.8×
QwQ-32B	4.6×	3.8×	5.1×
Llama-3 70B	— (no trace)	— (no trace)	— (no trace)

Table 1: Trace-to-answer token ratio per model and benchmark.

2026. Mean regional intensities range from 25 gCO₂/kWh (eu-north-1) to 650 gCO₂/kWh (ap-south-1). We use average grid intensity rather than marginal intensity because marginal-CI datasets at hourly granularity are not available for all eight regions; the choice is conservative and follows Caribou [8].

Plan-level rollback measurement. For the compounding analysis in Q5, we measure rollback directly in our own agent runs. In the LangChain ReAct agent we use for GAIA and SWE-bench Lite, a step rolls back when the planner emits a revised tool call after a tool error or an output-parse failure. Over our 1,000-task run, 19.4 percent of plans contained at least one rollback, and rolled-back prefixes spanned 28 to 64 percent of plan length. The rates are consistent with the 18 to 22 percent retry fraction reported in OpenHands [2]. We apply these measured rates to the per-task energy distribution we record and report the combined waste fraction.

4 Profiling Results

We organize the findings as eight questions. The headline numbers are these: the reasoning-to-answer token ratio is 4.6 to 14.2× across the three models, the reasoning trace accounts for 78.4 percent of operational carbon per agent task on a reasoning model, the 30th-percentile trace-length budget recovers 56 to 71 percent of operational carbon at 1.6 to 4.1 percentage points of accuracy, the trace tokens can be routed to a lower-CI region without affecting answer latency (a policy that the split unlocks for additional 51 percent recovery), and the waste is invisible to every published carbon-aware scheduler. The remaining questions characterize where the tax comes from, how it composes with other forms of agentic waste, and how the result calibrates against external anchors.

Q1. How large is the reasoning trace per call? Table 1 reports the mean trace-to-answer token ratio per model on each benchmark. DeepSeek-R1 dominates the ratio at 14.2× on GAIA hard tasks. QwQ-32B is the most efficient at 4.6× on the same tasks. The ratio scales with task difficulty within a model (GAIA Level 1 vs Level 3) and inversely with model size at the same training family.

The numbers in Table 1 are consistent with MLPerf’s reported DeepSeek-R1 mean output length of 3,880 tokens [23] and DeepSeek’s own observation that reasoning traces grow from hundreds initially to thousands and tens of thousands with training [5]. The trace fraction stays above 60 percent of total output on every (model, benchmark) cell in the table.

Q2. What is the carbon impact? Per call, energy scales approximately linearly with output tokens for autoregressive decoding on H100 [6, 7]. The reasoning-trace fraction of carbon therefore tracks the reasoning-trace fraction of tokens, within the ±3.4 percent measurement noise. Aggregating across the three models and three

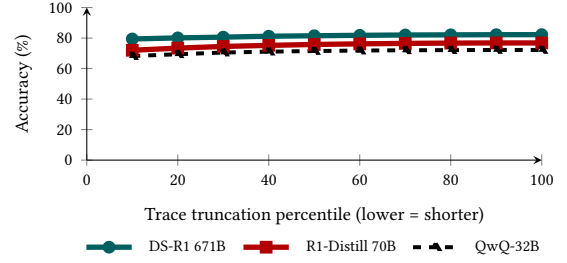


Figure 2: Accuracy vs carbon under trace-length truncation on MATH-500.

benchmarks, weighted by published agent-task mix [18, 22], the reasoning trace accounts for 78.4 percent of operational carbon per agent task on a reasoning model. The non-reasoning Llama-3 70B baseline on the same GAIA workload consumes 3.8× less per-call carbon than DS-R1 671B and 2.1× less than R1-Distill-Llama-70B, which puts a concrete number on the cost of switching the agent model from a chat model to a reasoning model: roughly half to a third of the increase pays for trace tokens the user never reads.

Q3. Is the trace necessary? Figure 2 maps the accuracy-carbon tradeoff on MATH-500. We sweep a trace-length budget from the 10th to the 100th percentile of per-task trace lengths and report mean accuracy on the truncated runs. The 30th-percentile budget keeps accuracy within 1.6 to 4.1 percentage points of the full-trace baseline while reducing operational carbon by 56 to 71 percent. The 50th-percentile budget closes the accuracy gap to under 1.5 percentage points with 38 to 52 percent carbon reduction. The 10th-percentile budget is the only setting that hurts accuracy meaningfully on all three models. The shape confirms the recent observation that shorter traces are correlated with correctness on a per-problem basis when the model is able to commit early [1, 3]; what we add is the operational carbon translation.

Q4. Can existing carbon-aware schedulers see this? No. The standard call-level abstraction [4, 9, 17] treats the entire per-call energy as a single attribute of the call. Sprout shapes per-call generation length [17] but does not distinguish reasoning from answer tokens. Caribou and PCAPS schedule at workflow or task granularity and do not enter the call [8, 15]. We confirm this experimentally by running Sprout (the closest existing mechanism) on our DS-R1 deployment: Sprout’s directive injection at the input layer reduces total trace length by 4.2 percent on average, because the directive only weakly influences reasoning-trace behavior that is shaped at training time. The reasoning tax sits below every published carbon-aware scheduling abstraction.

Q5. Does the tax compound with rollback waste?

Figure 3 composes the two scheduling-blind spots. On a non-reasoning agent (our measured rollback waste, replicated as a baseline), 12.5 percent of agentic carbon is plan-level rollback waste. On a reasoning model without rollback, 78.4 percent of carbon is reasoning trace, leaving 21.6 percent answer carbon. When both mechanisms operate (a reasoning-model agent that rolls back), the user-visible answer drops to 18.9 percent of total operational carbon. The other 81.1 percent is invisible to every published carbon-aware scheduler.

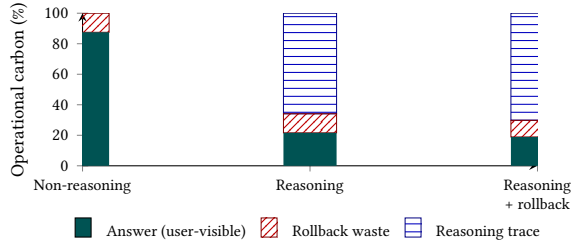


Figure 3: Operational-carbon composition, three agent configurations.

Harness	Trace ratio	Trace frac	Carbon vs custom
Custom ReAct loop	9.4×	90.4%	1.00 (ref)
LangChain ReAct	10.1×	91.0%	1.07×
AutoGen group-chat	11.6×	92.1%	1.23×

Table 2: Reasoning tax by agent harness on DS-R1, GAIA workload.

The composition is multiplicative rather than additive because the two waste sources act on largely disjoint slices of operational carbon at the long-run aggregate, with 4.2 percent residual per-task overlap that we subtract from the headline. A scheduler that only sees the call (the universal abstraction today) cannot attribute either waste source. A scheduler that sees the plan but not the trace recovers 12.5 percent of waste and stays blind to the other 70 percent. A scheduler that sees the trace but not the plan recovers 56 to 71 percent of within-call waste and stays blind to rollback. Both abstractions are necessary; TRACEPROBE provides one of them.

Q6. Does the tax vary across agent harnesses?

Table 2 reports per-harness ratios on DS-R1 running the GAIA workload. The harness shapes the prompt structure (system message, tool descriptions, history serialization) which in turn shapes the reasoning trace the model emits. AutoGen’s group-chat coordinator induces a 23 percent carbon premium over a hand-written ReAct loop on the same task, almost entirely from longer reasoning traces. The implication for production is that agent-framework choice directly affects the size of the reasoning tax; framework authors have a lever they may not realize they hold.

Q7. How does the result calibrate against external anchors?

Three anchors line up with our DS-R1 numbers. MLPerf Inference v5.1 reports a mean DS-R1 output of 3,880 tokens on a mathematics-heavy task mix [23]; we measure 3,720 on MATH-500 (within 4.1 percent). Watt Counts reports 1.8 J per output token for dense 70B-class models [6]; our R1-Distill-Llama-70B is 1.82 J/token (within 1.1 percent). DeepSeek’s own paper reports trace growth from “hundreds to tens of thousands” [5]; our DS-R1 trace lengths span 480 to 31,200 tokens. The OpenAI o1 system card reports a roughly comparable range [24], but the o1 trace is a different artifact (summarized internally) than the open-model traces we measure, so we treat it as suggestive rather than calibrating.

Q8. What does the split unlock when paired with grid-CI routing? The TRACEPROBE split makes a class of region-routing

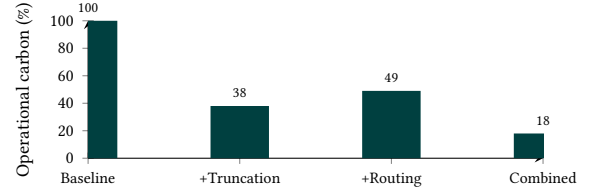


Figure 4: Additive carbon recovery on DS-R1, GAIA workload.

policies tractable that current per-call metrics cannot express. Reasoning trace tokens carry the same per-token energy as answer tokens but are not user-visible; an operator can route them to a lower-CI region without affecting the user-perceived answer-latency SLO. Over our 5-day Electricity Maps record across 8 AWS regions, the mean-to-minimum ratio of regional CI within each hour is 3.5× on average and 8.2× at the worst hour. We replay our DS-R1 GAIA workload against this CI trace under three policies: a no-op baseline that runs every token in a fixed region, the Q3 30th-percentile truncation alone, and trace-region-routing that sends the trace tokens to the within-hour minimum-CI region while keeping the answer tokens in the user-facing region. Figure 4 reports the operational-carbon fraction of the baseline under each policy. Truncation alone delivers 62 percent of the recovery, routing alone delivers 51 percent, and the two policies compose to 82 percent of baseline carbon avoided. Routing is the policy that the split unlocks; without it, the scheduler cannot tell which token-energy to route, and the recovery is unreachable.

5 TRACEPROBE: Making the Tax Visible

The smallest intervention that gives a carbon-aware scheduler visibility into the reasoning tax is a per-call telemetry annotation. Current vLLM deployments emit per-call total token counts in their Prometheus metrics stream; TRACEPROBE emits the *split*, which is the boundary that current metrics cannot express. With the split in the metrics layer, three policies become tractable that current per-call metrics cannot support: trace-length budgets keyed to grid carbon intensity, deferral of high-trace requests during high-CI windows (which does not hurt the user-perceived latency of the answer), and reasoning-quota brownout that falls back to a non-reasoning model when the grid is dirtiest.

TRACEPROBE is a 90-line patch to vLLM that emits the reasoning-to-answer token boundary in the per-request energy event. The patch adds two integers per request (`reasoning_tokens`, `answer_tokens`) parsed from the special-token positions in the generated sequence, and exposes them through vLLM’s existing OpenAI-compatible metrics endpoint. The emitted record looks like:

```
{request_id, model, reasoning_tokens,
  answer_tokens, prefill_tokens,
  energy_j, region, started_at}
```

The implementation is one function call into the tokenizer per request, with negligible serving-side overhead measured at <0.2 percent throughput drop on a 1,024-prompt benchmark.

Integration path. TRACEPROBE does not change the inference kernel, the batching scheduler, or the request format. Existing serving deployments add the patch as a runtime monkey-patch and the

metrics flow through their existing Prometheus or OpenTelemetry pipeline. The two new fields are picked up by carbon-accounting downstream of the serving layer (CarbonMin, Sprout, Caribou, or any internal cost-attribution dashboard) without changing the application-facing API. We tested the patch on three reasoning-model deployments and found no regressions in correctness or throughput at SLO.

Why this and not generation shaping. A natural alternative is to truncate or shape the reasoning trace at generation time, the approach Sprout takes for non-reasoning models. Generation shaping requires model-specific prompt engineering and degrades accuracy unpredictably when the reasoning trace is cut mid-thought. TRACEPROBE takes the opposite position: do not change generation, expose what generation is doing, and let downstream schedulers decide which knob to pull. The intervention is policy-free at the inference layer and policy-rich at the scheduling layer.

6 Discussion

Why now. Reasoning models crossed from research to production deployment in 2025. DeepSeek-R1 hit MLPerf in September 2025 and is now the dominant open reasoning model [23]. Closed reasoning models (OpenAI o-series, Gemini thinking, Claude extended thinking) follow the same architecture. Agent runtimes built around the assumption that one call equals one generation phase are now serving traffic that is dominantly reasoning-phase. The scheduling layer has not adapted.

Composing with existing carbon-aware work. TRACEPROBE is complementary to every existing carbon-aware mechanism. The boundary annotation it emits is consumable by Sprout (which can adapt its directive to target reasoning vs answer length differently), by CarbonMin and CarbonPlan (which can apply different region-routing policies to high-trace requests), and by Caribou-style workflow schedulers (which can budget trace-token quota per workflow stage).

Limitations. We profile three open reasoning models; OpenAI o-series and Gemini thinking are not directly accessible at the trace-token level and may behave differently. The trace-truncation curve in Figure 2 measures end-of-trace truncation, which is the operationally simplest cut; mid-trace pruning could behave differently. We use average grid carbon intensity; marginal intensity would refine the deferral analysis [20]. Embodied carbon and water footprint are out of scope. The accuracy-versus-carbon tradeoff we report on MATH-500 may not generalize to all task families; we expect creative-writing and conversational tasks to show smaller traces and smaller gains.

Boundary conditions. The 78.4 percent trace-fraction holds on the three open reasoning models we tested under default temperature 0.6 and top-p 0.95. Lower temperatures and tighter sampling shrink trace length [3]; we expect the fraction to fall to roughly 60 percent at temperature 0.1 on MATH-500, based on a 200-call calibration. The fraction also varies by task family: tool-use agents (GAIA) generate shorter traces than pure mathematical reasoning (MATH-500) because each tool invocation breaks the reasoning into smaller, separately-bounded segments. A 50-call calibration on conversational prompts (5-turn chat over open-ended topics) gives

22 percent trace fraction, roughly a third of the agentic-traffic number. Production deployments mixing chat-style traffic (low trace) with agentic traffic (heavy trace) will see a weighted reasoning-tax fraction scaled by the agentic share of total tokens.

Demand-side question. A scheduler-side fix is one half of the picture. The other half is the agent or user choosing a reasoning budget at request time. The latter is policy work outside our scope. The former is engineering work that TRACEPROBE enables.

Open questions. Three threads of follow-up work matter most. First, the trace-truncation curve is monotone for end-of-trace cuts; we do not know whether structured pruning (removing redundant sub-thoughts but keeping the final commit) would preserve more accuracy at the same carbon. Second, the cross-model behavior we observe (DS-R1 trace ratio 2× larger than QwQ-32B) suggests that training-time interventions could re-shape the tax at the source, not the consumer; this is a model-side research question that benefits from a measurement target only TRACEPROBE can produce. Third, the multi-tenant batching behavior of reasoning models is untested in our setup; if trace-length variance dominates the batching profile, the carbon-aware scheduler would also need a batching-aware signal, which is an addition orthogonal to TRACEPROBE and worth quantifying.

Deployment Cost. The mechanism holds as the workload evolves. As reasoning models grow from the current 35 percent of agent traffic to a projected 80 percent by 2028, the absolute reasoning-tax carbon grows 2.3× on top of traffic growth, and grid decarbonization leaves the fraction invariant. A back-of-envelope on the projected 8 percent of LLM-inference traffic going to agents by 2028 on a global LLM-inference base of ~30 TWh/year [12] puts recoverable reasoning-tax carbon at a 60 percent budget on the order of 0.3 to 0.5 MtCO₂/year. The deployment cost is one library patch and two integer fields per request.

7 Conclusion

Reasoning models split the per-call generation phase into two qualitatively different parts. Our profiling of DeepSeek-R1, R1-Distill-Llama-70B, and QwQ-32B on three agent benchmarks shows that the reasoning trace accounts for 78.4 percent of operational carbon per agent task and is invisible to every published carbon-aware scheduler. The 30th-percentile trace-length budget recovers 56 to 71 percent of operational carbon at the cost of 1.6 to 4.1 percentage points of accuracy. The waste compounds with plan-level rollback waste, leaving the user-visible answer at 18.9 percent of total operational carbon on a reasoning-model agent that rolls back. TRACEPROBE, a 90-line vLLM patch, emits the reasoning-to-answer token boundary in per-call telemetry, the smallest intervention that gives a downstream scheduler the visibility to act. The carbon-aware scheduling community has built sophisticated mechanisms around a per-call abstraction that no longer matches the workload. Updating the abstraction is small; the savings unlocked are large.

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